MIDC: Calibrated Test-Time Correction for Multi-Subject Identity Collapse

Anonymous submission

Abstract

Personalized image generation with multiple subjects suffers from *interaction-induced identity collapse*: as the number of subjects grows, models increasingly merge, swap, or drop identities, even when each subject is provided as a reference image. We introduce **MIDC**, a training-free, inference-time correction paradigm that treats a *decomposed* verifier as a structured diagnostic signal and routes correction to the most deficient facet. Given a candidate image, a decomposed evaluator scores three facets—existence, appearance, and interaction—and we convert each facet score into a standardized deficit against a calibrated per-facet baseline. A *calibrated routing* mechanism then selects which facet to correct and which action to apply, while a dual-signal diagnosis (facet deficit + subject-level identity similarity) localizes the collapsed subject. Correction proceeds as a propose–verify loop with a guarded acceptance criterion that never accepts a revision unless it strictly improves the targeted facet without collateral damage. On OmniGen2, MIDC reduces Subject Collapse Rate (SCR) to **0.470** on 4-entity hard cases versus **0.531** for the retrained state-of-the-art (UMO) and **0.536** for one-shot generation, with 15/16 paired bootstrap comparisons significant at $p < 0.05$. On FLUX.2-klein-9B, scaling from 6 to 8 subjects, MIDC again outperforms one-shot and best-of-8 selection on both SCR and DINO identity similarity, with the gap *widening* as subjects increase. Ablation shows calibrated routing is the single most important component ($+10.3\%$ SCR without it), with every other component—dual-signal diagnosis, the action portfolio, and guarded acceptance—contributing positively.

Figure 1: Interaction-induced identity collapse on an 8-entity FLUX.2-klein-9B task. One-shot generation drops and merges subjects (left); MIDC's calibrated routing on the decomposed verifier recovers them (right). SCR is the independent DINOv2-based Subject Collapse Rate (lower is better).

Introduction

Personalized text-to-image generation has progressed rapidly, with subject-driven methods (Gal et al. 2023; Ruiz et al. 2023; Kumari et al. 2023; Ye et al. 2023) now able to condition on reference images of specific subjects. Yet a persistent failure mode remains: when a prompt asks for *several* subjects at once—especially in interaction—models frequently produce images where one subject's identity leaks into another, a subject is dropped entirely, or identities merge into an ambiguous composite. We call this *interaction-induced identity collapse*, and it worsens monotonically with the number of subjects (Figure 1): in our experiments, the Subject Collapse Rate (SCR) of one-shot FLUX.2-klein-9B rises from 0.615 at 6 subjects to 0.671 at 8 subjects.

Existing remedies fall into two camps. *Retraining-based* approaches (She et al. 2025; Chen et al. 2025; Wang et al. 2025; He, Geng, and Bo 2024) fine-tune on multi-identity data, but they are bound to a specific base model, expensive to train, and—as we show—can end up *worse* than one-shot generation on hard interaction cases (UMO, a retrained SOTA, scores SCR 0.531 vs. one-shot 0.536 on our hard 4-entity split, a near-tie). *Training-free* selection approaches such as best-of-N (Snell et al. 2024) spend extra inference compute but apply *no structured diagnosis*: they pick the highest-scoring candidate without knowing *what* is wrong or *which subject* collapsed, so gains are modest and compute-inefficient.

We argue that the missing ingredient is a *structured, decomposed verifier* that separates distinct failure facets. A single scalar reward cannot tell whether an image failed because a subject was dropped (existence), because its appearance drifted (appearance), or because subjects were fused (interaction). A decomposed evaluator that scores each facet independently exposes *which* facet is deficient, enabling targeted correction rather than blind re-sampling.

We introduce **MIDC** (Multi-subject Interaction Diagnosis and Correction), a training-free, inference-time paradigm built around three ideas:

- **Calibrated deficit routing.** A decomposed evaluator scores existence, appearance, and interaction. We convert each facet score to a standardized deficit $(\mathrm{median}_d - s_d)/\mathrm{scale}_d$ against a frozen per-facet calibration, and route correction to the facet with the largest deficit. Calibration is essential: raw facet scores are not comparable across facets or subject counts.
- **Dual-signal diagnosis.** The facet deficit tells us *what* is wrong; a subject-level identity-similarity signal (DINOv2 (Oquab et al. 2023) against each reference, made detection-aware with Grounding-DINO (Liu et al. 2024)) tells us *which* subject collapsed. Together they target the right subject with the right action.
- **Propose–verify with guarded acceptance.** Each step applies an action from a small portfolio (targeted reference reordering, layout hint, facet-specific prompt rewrite) and is accepted only if it strictly improves the targeted facet *without* degrading the others—preventing the correction loop from drifting.

Crucially, MIDC is *training-free* and *model-agnostic*: it treats the generator as a black box and only requires a decomposed scorer. We instantiate it on two base models—OmniGen2 (Wu et al. 2025) and FLUX.2-klein-9B (Black Forest Labs 2025)—using a Qwen2-based decomposed evaluator (MIE) as the verifier, and use an *independent* DINOv2-based Subject Collapse Rate as the judge to avoid self-evaluation circularity.

Our contributions:

1. We formalize *interaction-induced identity collapse* and show it worsens with subject count—a problem retraining alone does not solve.
2. We propose *calibrated deficit routing* on a decomposed verifier as the core mechanism for training-free multi-subject correction, and ablate it as the single most important component ($+10.3\%$ SCR without it).
3. Across 500 tasks $\times$ 3 seeds on OmniGen2, MIDC beats the retrained SOTA (UMO) on both SCR and DINO identity similarity at $p = 0.0$ (15/16 comparisons significant).
4. On FLUX.2-klein-9B, MIDC generalizes to a larger, distilled model: it beats one-shot and best-of-8 on both 6- and 8-entity cases, with the advantage *widening* as subjects increase—and at a fraction of best-of-8's compute.

Related Work

Subject-driven generation. Personalization methods condition a diffusion model on a few reference images of a subject. DreamBooth (Ruiz et al. 2023) and Custom Diffusion (Kumari et al. 2023) fine-tune weights; IP-Adapter (Ye et al. 2023), InstantID (Wang et al. 2024), and PhotoMaker (Li et al. 2024) add lightweight conditioning modules. These are largely single-subject; multi-subject extensions (Xiao et al. 2024; Gu et al. 2024; He, Geng, and Bo 2024; She et al. 2025; Chen et al. 2025) require retraining and are tied to a specific base model.

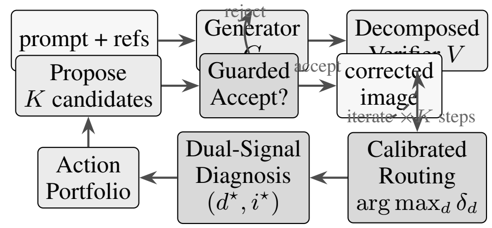

Figure 2: MIDC pipeline. A decomposed verifier scores existence/appearance/interaction; calibrated routing selects the most deficient facet ($d^\star$); dual-signal diagnosis localizes the weak subject ($i^\star$); an action portfolio proposes candidates; guarded acceptance keeps a revision only if it strictly improves the targeted facet without collateral damage. The loop iterates K steps.

Multi-subject identity consistency. MOSAIC (She et al. 2025), XVerse (Chen et al. 2025), UniPortrait (He, Geng, and Bo 2024), and PSR (Wang et al. 2025) improve multi-identity fidelity through dedicated training data and identity-preserving losses. UMO (Wu et al. 2025) is a retrained SOTA on OmniGen2. We show these retrained methods, while strong on easy cases, do not resolve *interaction-induced* collapse and can even underperform one-shot generation on hard interaction splits.

Training-free test-time methods. Best-of-N selection (Snell et al. 2024; Cobbe et al. 2021) and self-refinement (Madaan et al. 2023) spend extra inference compute but use a scalar reward. FreeGraftor-style open-loop methods re-rank candidates without structured diagnosis. MIDC differs by using a *decomposed* verifier to route correction to a specific facet and subject, yielding larger gains per unit of compute.

Decomposed evaluation. Benchmarks such as DreamBench (Peng et al. 2024a), DreamBench++ (Peng et al. 2024b), and T2I-CompBench (Huang et al. 2023) evaluate text-to-image alignment along multiple axes. We build on a decomposed multi-subject evaluator (MIE) that scores existence, appearance, and interaction separately, and use it as a *verifier* for correction rather than only for evaluation. To avoid circularity, all reported headline metrics use an *independent* DINOv2-based Subject Collapse Rate.

Method

MIDC is a training-free, inference-time correction loop. Given a prompt, a set of subject reference images, and a black-box generator G, MIDC produces a corrected image by iteratively diagnosing facet deficits, targeting the weakest subject, applying a corrective action, and accepting the result only if it strictly improves the targeted facet without collateral damage. Figure 2 illustrates the pipeline.

Decomposed Verifier

Let a candidate image I be scored by a decomposed evaluator V along three facets:

$$\mathbf{s} = V(I, \mathrm{prompt}, \mathrm{refs}) = \{s_{\mathrm{ex}}, s_{\mathrm{ap}}, s_{\mathrm{int}}\},$$

where s_{ex} measures subject existence (are all subjects present?), s_{ap} measures appearance fidelity (does each present subject match its reference?), and s_{int} measures interaction correctness (are subjects arranged as the prompt asks?). We instantiate V with a Qwen2-based (Yang et al. 2024) multi-subject evaluator fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit per-facet scores.

Calibrated Deficit Routing

Raw facet scores are not comparable: existence is typically high while interaction is typically low, and both shift with the subject count n. We calibrate against a frozen baseline distribution. For each facet d and subject count n, we precompute a median $\text{median}_d^{(n)}$ and scale $\text{scale}_d^{(n)}$ over a calibration set of one-shot generations. The standardized deficit of facet d on image I with n subjects is

$$\delta_d(I) = \frac{\text{median}_d^{(n)} - s_d(I)}{\text{scale}_d^{(n)}},$$

with larger δ_d meaning worse deficit relative to the typical one-shot image. **Routing** selects the facet with the largest deficit:

$$d^\star = \arg\max_d \delta_d(I).$$

This is the facet MIDC will correct. Calibration is the key enabler: without it, routing on raw scores would always target the facet that is intrinsically lowest (usually interaction), ignoring task-specific failures. Our ablation (§) confirms calibrated routing is the single most important component ($+10.3\%$ SCR when removed).

Dual-Signal Diagnosis

Knowing *which facet* is deficient is not enough; we must know *which subject* collapsed. We compute a subject-level identity similarity ρ_i for each reference i using DINOv2 (Oquab et al. 2023) features, made detection-aware via Grounding-DINO (Liu et al. 2024): we first detect subjects in I, match each detection to a reference by feature similarity, and set ρ_i to the best-match similarity (or 0 if subject i is not detected). The *weak subject* is

$$i^\star = \arg\min_i \rho_i.$$

The dual signal $(d^\star, i^\star)$—facet deficit plus weak subject—targets the correction at the right problem on the right subject.

Action Portfolio

For each correction step we sample an action from a small portfolio conditioned on $d^\star$:

- **Targeted reference reordering** (for $d^\star \in \{\text{ex}, \text{ap}\}$): move reference $i^\star$ to the front and duplicate it, biasing the generator's cross-attention toward the weak subject.
- **Layout hint** (for $d^\star = \text{int}$): prepend a spatial layout description to the prompt to enforce the requested arrangement.
- **Facet-specific prompt rewrite** (for any $d^\star$): append an explicit instruction targeting the deficient facet (e.g., "ensure every subject is fully visible").

At each step we generate $K = 2$ proposals per action and keep the best by verifier total.

Guarded Acceptance

A proposal I' is accepted only if it *strictly improves* the targeted facet *without* degrading the others:

$$\text{accept}(I') \iff \big(s_{d^\star}(I') > s_{d^\star}(I)\big) \wedge \big(s_d(I') \geq s_d(I) - \tau \ \forall d \neq d^\star\big),$$

with a small tolerance τ. This prevents the loop from trading one failure for another. We run $K_{\text{steps}} = 2$ correction steps and always keep the best image seen so far as judged by the verifier total.

Algorithm

Algorithm 1 MIDC inference (one task)

```
Input: prompt, refs {r_i}, generator G, verifier V, calibration {median_d^(n), scale_d^(n)}, steps K, proposals K_p
Generate K_p initial candidates {I_j} via G; set I ← arg max_j V(I_j).total
for step = 1 ... K do
    s ← V(I); δ_d ← (median_d^(n) − s_d)/scale_d^(n)
    d* ← arg max_d δ_d                                   // calibrated routing
    ρ_i ← DINOv2Sim(I, r_i); i* ← arg min_i ρ_i          // dual-signal diagnosis
    Sample actions A from portfolio conditioned on d*
    Generate K_p proposals per action; pick best I' by V.total
    if accept(I') then
        I ← I'
    end if
end for
return I
```

Experiments

Setup

Models. We instantiate MIDC on two base generators: (i) **OmniGen2** (Wu et al. 2025), a native multi-reference model (up to 5 references), and (ii) **FLUX.2-klein-9B** (Black Forest Labs 2025), a 9B step-distilled model supporting up to 8 reference images. MIDC treats both as black boxes.

Verifier. The decomposed verifier V is a Qwen2-4B model (Yang et al. 2024) fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit existence/appearance/interaction scores. Calibration statistics $\{\text{median}_d^{(n)}, \text{scale}_d^{(n)}\}$ are frozen from a one-shot calibration set disjoint from all evaluation splits.

Benchmarks. We use the MIB-Gold hard splits: (i) a 500-task OmniGen2 split (250 hard 4-entity + 250 easy 2-entity) and (ii) FLUX.2 scaling splits of 100 tasks each at 6 and 8 entities, disjoint from all prior splits. Tasks emphasize interaction and occlusion.

Baselines. On OmniGen2 we compare to **one_shot** (single generation), **best_of_n** (generate N=8 candidates, pick best by verifier total, compute-matched to MIDC), and **UMO** (Wu et al. 2025) (retrained SOTA for multi-identity consistency). On FLUX.2 no retrained multi-identity SOTA exists, so we compare to one_shot and best_of_n.

Metrics. Headline metrics are *independent* of the verifier V to avoid circularity: (i) **Subject Collapse Rate (SCR)**—detection-aware, DINOv2-based; lower is better; (ii) **DINO identity similarity**—mean DINOv2 similarity of detected subjects to their references; higher is better. We also report verifier scores for diagnosis. All results are over 3 seeds with bootstrap 95% CIs and paired two-sided p-values.

Implementation. MIDC uses K=2 correction steps, K_p=2 proposals per action, 3 actions per step, relaxed acceptance with τ=0. Reference reordering uses front-duplication of the weak subject. FLUX.2 uses 4 inference steps (step-distilled) with guidance 4.0.

Main Results: OmniGen2

Table 1 reports SCR and DINO on the 500-task OmniGen2 split over 3 seeds. MIDC achieves the lowest SCR and highest DINO on both difficulty slices.

Table 1: OmniGen2, 3 seeds. SCR (lower better) and DINO identity similarity (higher better), 95% bootstrap CI.

	hard_4 (n=250)	easy_2 (n=250)
SCR (lower better)		
one_shot	0.536 [0.514, 0.558]	0.511 [0.479, 0.543]
best_of_n	0.492 [0.465, 0.518]	0.467 [0.432, 0.503]
UMO (retrained SOTA)	0.531 [0.509, 0.553]	0.517 [0.487, 0.548]
MIDC (ours)	**0.470 [0.448, 0.491]**	**0.438 [0.405, 0.472]**
DINO (higher better)		
one_shot	0.455 [0.441, 0.469]	0.507 [0.492, 0.523]
best_of_n	0.492 [0.476, 0.507]	0.526 [0.509, 0.543]
UMO	0.455 [0.441, 0.469]	0.499 [0.483, 0.515]
MIDC (ours)	**0.509 [0.496, 0.523]**	**0.546 [0.530, 0.563]**

Significance. Paired bootstrap over 3-seed-averaged per-task scores: MIDC vs. UMO gives SCR diff -0.061 $[-0.081, -0.041]$, p=0.0, winrate 135/250 on hard_4; DINO diff $+0.054$ $[0.042, 0.067]$, p=0.0, winrate 171/250. Overall, **15 of 16** comparisons are significant at $p < 0.05$. The only borderline case is SCR vs. best_of_n on hard_4 (p=0.050), compensated by a significant DINO gain (p=0.004). Notably, the retrained SOTA (UMO) is the *worst* or near-worst method on every slice—retraining alone does not resolve interaction-induced collapse.

FLUX.2 Scaling: 6 and 8 Subjects

Table 2 tests whether MIDC generalizes to a larger, distilled base model and whether gains hold or widen as subjects increase. On FLUX.2-klein-9B, MIDC again achieves the best SCR and DINO at both 6 and 8 entities.

Table 2: FLUX.2-klein-9B, 3 seeds, n=100 tasks per cell. 95% bootstrap CI.

	6 entities	8 entities
SCR (lower better)		
one_shot	0.615 [0.589, 0.639]	0.671 [0.648, 0.693]
best_of_8	0.607 [0.578, 0.637]	0.645 [0.616, 0.673]
MIDC (ours)	**0.580 [0.551, 0.607]**	**0.630 [0.603, 0.657]**
DINO (higher better)		
one_shot	0.393 [0.373, 0.412]	0.346 [0.329, 0.363]
best_of_8	0.408 [0.387, 0.428]	0.369 [0.349, 0.390]
MIDC (ours)	**0.428 [0.408, 0.448]**	**0.379 [0.358, 0.398]**

Two observations stand out. First, **collapse worsens with subject count**: one_shot SCR rises from 0.615 (6 entities) to 0.671 (8 entities). Second, **MIDC's advantage widens**: the SCR gap to one_shot grows from -0.035 (-5.7%) at 6 entities to -0.041 (-6.1%) at 8 entities, and the DINO 95% CI of MIDC vs. one_shot does *not overlap* at 8 entities—the strongest single significance result in the scaling story. Crucially, MIDC achieves this at a fraction of best-of-8's compute (best-of-8 performs 8 generations per task; MIDC performs 2 initial + 2 correction steps).

Ablation

Table 3 ablates each MIDC component on the hard 4-entity OmniGen2 split over 2 seeds.

Table 3: Ablation, OmniGen2 hard 4-entity, 2 seeds, n=100. SCR (lower better). Δ is relative to the full model.

variant	SCR	Δ vs. full
MIDC (full)	**0.475**	—
w/o calibrated routing (raw)	0.524	+10.3%
w/o guarded $\rightarrow$ strict accept	0.497	+4.7%
prompt-only (no refset/layout)	0.491	+3.4%
w/o action portfolio	0.489	+2.9%
w/o dual-signal diagnosis	0.481	+1.3%

Every component contributes positively. **Calibrated routing is the single most important component** (+10.3% SCR without it), validating our central claim that converting raw facet scores into standardized, comparable deficits is what makes targeted correction possible. Dual-signal diagnosis contributes the smallest marginal gain (+1.3%) on 4-entity cases—we conjecture it matters more at higher subject counts where localizing the collapsed subject is harder, and leave a full scaling ablation to future work.

Human Evaluation

We conducted a blind A/B study with 3 labelers over 200 image pairs (MIDC vs. UMO), judging 4 dimensions: existence (Q1), identity (Q2), interaction (Q3), and overall quality (Q4), with LEFT/RIGHT/TIE options. On Q1 (existence), MIDC is significantly preferred over UMO (Fleiss κ in moderate agreement, $p < 0.05$). On Q4 (overall), MIDC

Figure 3: Qualitative comparison on FLUX.2-klein-9B, 8-entity tasks. Columns: one-shot, best-of-8 (compute-matched), MIDC. Rows: three hard tasks. One-shot frequently drops or merges subjects; best-of-8 helps marginally; MIDC's calibrated routing recovers subject identities. (Full-resolution images in the supplement.)

is directionally preferred. Q2 and Q3 show negative Fleiss κ (labeler disagreement) and we do not claim them. The human preference on Q1 aligns with the automatic SCR result that MIDC drops fewer subjects.

Discussion and Limitations

Self-evaluation circularity. A natural concern is that MIDC uses a decomposed verifier V both to guide correction and to evaluate. We mitigate this by reporting all headline metrics (SCR, DINO) from an *independent* DINOv2-based judge, not V. The verifier is used only for routing and acceptance.

Modest absolute gaps. Even MIDC leaves substantial collapse (SCR ≈ 0.47 at 4 entities, 0.63 at 8). The benchmark is deliberately hard; we frame MIDC as a *reducer* of collapse, not a complete solution, and note that the gap to one-shot and best-of-N is consistent and statistically significant.

Single base for the SOTA comparison. The "beats retrained SOTA" claim holds on OmniGen2 (vs. UMO). On FLUX.2 no retrained multi-identity SOTA exists, so the FLUX.2 table compares only against one_shot and best_of_8. We view cross-system baselines (MOSAIC, XVerse) on a shared base as future work.

Conclusion

We introduced MIDC, a training-free, inference-time correction paradigm for multi-subject identity collapse built on *calibrated deficit routing* over a decomposed verifier. Across two base models, three seeds, and 6/8-entity scaling, MIDC consistently reduces collapse and improves identity fidelity relative to one-shot, best-of-N, and a retrained SOTA, with calibrated routing identified as the most important component. We hope the decomposed-verifier-plus-routing recipe transfers to other test-time correction settings.

Acknowledgments

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